
Modeling Emergent Phenomena with Transformer Cellular Automata

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Abstract

Emergence is a mystery at the heart of many human endeavors. How do particles become molecules? How do molecules become cells? How do cells become beings? We propose an architecture called transformer cellular automata (TCA) for learning models of complex systems based on partial observations. Inspired by the multi-modal applicability of transformers and recent work on neural cellular automata (NCA), we aim for a robust, expressive, and generalizable model of emergent phenomena. We propose the application of TCA to a chaotic active matter fluid system to demonstrate these capabilities and improve our understanding of this important model system.

1. Introduction

Emergence, the spontaneous formation of complex systems from simple components and simple interactions, is a captivating phenomenon driving life’s existence and the universe’s richness (O’Connor, 2021). Yet, understanding it remains a scientific challenge, as traditional models often fall short in encapsulating its complexity (Kalantari et al., 2020). We propose a novel computational model - the Transformer Cellular Automata (TCA). Drawing from the strengths of transformer neural networks and cellular automata, TCA aims to infer the dynamics of emergent phenomena efficiently, even when only partial observations are available.

The value of a strong general model for inferring the dynamics of a complex system is difficult to understate. Strong models for complex dynamics are important for enhancing experimental throughput (i.e., it’s easier to simulate 1000 experiments that perform them by hand). They are also valuable for informing theory, both through higher *in silico* experimental throughput and through direct analysis of the models. Computational models also allow for greater

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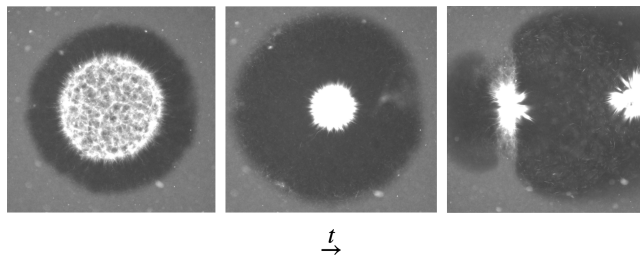


Figure 1. Light activated active matter microfluidics experiment, courtesy of Shichen Liu of (Yang et al., 2022)

observability, in general, than real experiments. Of course, this comes at the cost of approximation error, which further motivates research on strong models that have low approximation error. It is our profound hope that this work may also aid in the generation of cohesive theory on complex systems – something which is sorely lacking in the current literature on the subject.

We discuss related work, the specifics of the TCA design, and proposals for applying TCA to modeling the dynamics of a complex light-activated biophysical active matter system (Figure 1).

2. Related Work

2.1. Neural Cellular Automata

Cellular automata (CA) are a classic model of computation composed of a grid of cells, each having one of a finite set of states at a given time. The states are updated based on a rule set mapping the current state of a cell and its immediate neighboring cells to a new value for the cell (Weisstein). Importantly, the same rule is applied to update every cell. They are an influential model of complex systems in mathematics, computer science, physics, and beyond.

Neural cellular automata (NCA) allow for non-finite states (e.g., real vector valued states) and a neural network to parameterize the update rule. The paper “Growing Neural Cellular Automata” by Mordvintsev et al. (Mordvintsev et al., 2020) tackles the daunting challenge of computationally modeling morphogenesis, the process through which an organism develops its shape. If every cell has the same

“source code”, how do they know where to move, how to change, and how to signal their neighbors in order to actualize a complex large-scale organism?

Specifying a ruleset for cells to follow to achieve large-scale goals like morphogenesis is a daunting task. So, why not have a machine infer the rules via standard gradient-based optimization?

Each cell has a 16-dimensional state vector and the neighborhood is defined as a square lattice. There is a 50% chance that the neural network update rule will be applied to a given cell in a given iteration. When it does, the state of the cell and information about its immediate neighbourhood is input to the neural network update rule. The output of the neural network update rule is added to the state vector of the cell, thus propagating information and computing. This produces a fascinating “cell-like” computational structure, parameterized by a neural network that is optimizable with differentiable objectives.

By applying backpropagation through time (BPTT), the authors were able to train a single, small neural network update rule that robustly formed large-scale patterns, even in spite of perturbations. The relative ease with which such systems can be trained using modern deep learning methods is a great step forward for modeling complex emergent phenomena. You can witness this computational marvel in action in [this interactive demo](#).

2.2. Transformers

Originally proposed for tasks in Natural Language Processing, transformer blocks (Vaswani et al., 2017) have far exceeded their initial domain, proving competitive in a broad range of modalities (Akbari et al., 2021), including vision tasks (Arnab et al., 2021; Atito; Fan et al., 2021; Dosovitskiy et al., 2021). At their core, transformers implement an all-to-all mapping between elements of their input to compute an output, often of the same dimension as the input. Most importantly, this all-to-all mapping is **learnable**, allowing the model to “attend” to various parts of its input when the need arises. This key feature allows them to model complex dependencies within data, making them a promising tool for understanding complex systems.

Just as important is their ability to infer high-dimensional structure from partial observations. This ability is exemplified in their ability to learn the *structure* of language just by observing a large volume of text (Radford et al.; Brown et al., 2020). Similar structure-learning ability arises in other modalities, such as vision as mentioned above. This ability is of great interest for inferring the nature of complex systems and the interaction between elements, particularly when the state is only partially visible and “evidence” ought to be accumulated over time to inform subsequent predic-

tions.

2.3. Transformers for Learning Dynamics

Work such as “Learning stochastic dynamics and predicting emergent behavior using transformers” (cas) apply transformers to modeling the dynamics of complex systems. However, the work fails to take advantage of passing latent variables between time steps. Instead, they learn a mapping from state s_t to s_{t+1} of the system of interest using a transformer. This is a common trend observed in the work to date on applying transformers to learning system dynamics (Yan et al., 2022a; Geneva & Zabaras, 2020).

The approach is lacking when there are only partial observations of the system. In the active matter system, for instance, each particle’s state is not always visible. It is valuable to aggregate evidence from many preceding time steps and pass some **latent variable** through time to inform future activity.

This is where the insight from the neural cellular automata paper (Mordvintsev et al., 2020) comes to mind: what if we applied the same backpropagation through time training scheme to learn an update rule for a **transformer** cellular automata? As mentioned above, one can view NCA as a sort of recurrent CNN with a very specific kernel to aggregate information to inform state updates. What would a recurrent dynamical transformer look like? Instead of a regular square grid of cells, what if the “connectivity” of cells was learned via an attention mechanism? And the hidden state of each cell was propagated forward through time to aggregate information? This exciting idea is what we propose to explore in this work. To the best of the author’s knowledge, such a “transformer cellular automata” has been proposed only in (Teshfaldet et al., 2022) as an internal mechanism for a ViT-inspired vision denoising model.

Discussion of active matter modeling to date is covered in the discussion section for brevity.

3. Methods

Here we propose the TCA architecture and training scheme, based on transformer self-attention blocks described in (Vaswani et al., 2017) and neural cellular automata described in (Mordvintsev et al., 2020). While the iterative self-attention is similar to (Teshfaldet et al., 2022), our input formulation and training method differs substantially.

3.1. Autonomous TCA

We will begin by describing an autonomous form of the TCA architecture. Incorporating input (e.g., system observations) during runtime will be covered in the next subsection.

Let us represent the TCA model as TCA , a stack of

n_{blocks} self-attention blocks mapping from input tokens $\{x_i(t)\}_{i=1}^N$ of dimension d to output tokens $\{x'_i(t)\}_{i=1}^N$ of the same dimension. For simplicity, we use discrete time $t = 0, 1, \dots, T$. As the notation implies, the token set is transformed over time as:

$$\mathbf{X}(t+1) = \mathbf{X}(t) + TCA(\mathbf{X}(t))dt \quad (1)$$

where $\mathbf{X}(t)$ is a matrix of column vector tokens $\{x_i(t)\}_{i=1}^N$ and dt is a small constant.

Optionally, we can set the output dimension of $TCA(\cdot)$ to be less than d so that certain indices of each $x_i(t)$ are unaffected by the dynamics. These may include things like particle identity or other fixed information. The information in the tokens $x_i(t)$ thus has a very general quality. One can place **positional** information in the tokens as positional codes, allocate certain indices to observable state information and others to latent information like in (Mordvintsev et al., 2020). We are also free to remove indices of $x_i(t)$ from the input to the TCA .

Of particular interest is windowing of interactions. To limit information flow between distant indices, one may apply a **gated distance matrix** attention mask to the self-attention matrices. For tokens that are too far, the attention between them can be set to zero, whereas tokens within a defined distance may have non-zero attention values between them. This is an inviting way to realize physical information transmission limitations and reduce the quadratic time complexity of the self attention operation (Keles et al., 2022).

Training: The autonomous TCA can be trained to minimize any computationally tractable loss function defined on its state evolution $\langle \mathbf{X}(0), \mathbf{X}(1), \dots, \mathbf{X}(T) \rangle$. This includes training objectives like in (Mordvintsev et al., 2020) where might aim to minimize $\mathbb{E}_{\mathbf{X}(T)|\mathbf{X}^*(0)}[\|\mathbf{X}(T) - \mathbf{X}^*(T)\|]$.

3.2. Non-Autonomous TCA

Let us introduce some time-dependent external system $\mathcal{S}(t)$ from which we take measurements $\mathbf{S}(t) = [s_i(t)]_{i=1}^M$ is a matrix of M column vectors $s_i(t)$ with each $s_i(t) \in \mathbb{R}^{d_s}$. For a general external system with general measurements (i.e., arbitrary dimensionality d_s and number M), we introduce the following operators to facilitate the construction and training of the non-autonomous TCA:

1. **Encoder Map** $\zeta : \mathbb{R}^{d_s \times M} \times \mathbb{R}^{d \times N} \rightarrow \mathbb{R}^{d_v \times N}$
2. **Decoder Map** $\psi : \mathbb{R}^{d \times N} \rightarrow \mathbb{R}^{d_s \times M}$

Note that for systems with a 1-to-1 correspondence between TCA tokens x_i and observations $s_i(t)$ from the system, these maps can well be identity maps or a subset of the identity map.

The encoder map can be used to transform observations $\mathbf{S}(t)$ from system $\mathcal{S}(t)$ along with the current state of the TCA $\mathbf{X}(t)$ into a matrix that can be concatenated with $\mathbf{X}(t)$ as the input to the non-autonomous TCA with input token dimensionality $d + d_v$. Put concretely, the forward process described for the autonomous TCA in Equation 1 is now

$$\mathbf{X}(t+1) = \mathbf{X}(t) + TCA \left(\left[\begin{array}{c} \mathbf{X}(t) \\ \zeta(\mathbf{S}(t), \mathbf{X}(t)) \end{array} \right] \right) \quad (2)$$

and we can obtain the TCA's estimate the next observation from $\mathcal{S}$ as $\hat{\mathbf{S}}(t)$ via

$$\hat{\mathbf{S}}(t) = \psi(\mathbf{X}(t)) \quad (3)$$

Training: The non-autonomous TCA can be trained to minimize any computationally tractable differentiable loss function defined on its state evolution $\langle \mathbf{X}(0), \mathbf{X}(1), \dots, \mathbf{X}(T) \rangle$, which includes functions on the observation predictor model $\hat{\mathbf{S}}(t)$ such as

$$\mathcal{L}_{pred} = \mathbb{E}_{t, \mathcal{S}} \|\hat{\mathbf{S}}(t) - \mathbf{S}(t)\| \quad (4)$$

4. Experiments

4.1. Synthetic Data Baseline

We will first attempt to understand the utility of the autonomous TCA via a physics-based active matter model called *aLENS* (a Living Ensemble Simulator), presented in (Yan et al., 2022b), which simulates an active matter system of molecular motors and microtubules based on a thermodynamic energy landscape model with hard-body repulsion between filaments. *aLENS* is capable of simulating a system with $\sim 10,000$ components at the molecular level – giving direct access to ground truth particle trajectories to which we can compare our autonomous TCA's state evolution $\langle \mathbf{X}(0) \dots \mathbf{X}(T) \rangle$.

Deep learning models have yet to truly shine in the field of active matter, so we would be interested in understanding the efficacy of TCA as compared to less sophisticated systems like *Cytosim* (Nedelec & Foethke, 2007), a coarser but more computationally affordable model of the same type of active matter system where filaments and motors are modeled as points. We are also interested in comparing the performance of autonomous vs. non-autonomous TCA with varying degrees of information flow through the encoder map ζ for the non-autonomous system. Since there is a 1-to-1 correspondence, ζ could be defined as the identity map with random masking on the input (i.e., dropout). Loss curves on comparing $\mathcal{L}_{pred}$ from Equation 4 would be

valuable for understanding the performance characteristics of TCA. It would also be informative to compare TCA’s performance with that of Markovian models like the ones discussed earlier that map from the system observations at t to an estimate of the same at $t + 1$. Prior unpublished work in the Thomson lab have applied CNN-like architectures to this problem with mixed success, for instance. Both aLENS and Cytosim are open-source projects per (Nedelec & Foethke, 2007; Yan et al., 2022b).

4.2. Fitting Experimental Data

We are also interested in applying the non-autonomous TCA to experimental data collected from microfluidics experiments with the active matter system. I have secured novel experimental data collected using the apparatus from (Yang et al., 2022). It consists of 36 high-quality (2048×2048 pixels) recordings with durations between 8 and 10 seconds each imaged with bright-field microscopy.

An important challenge associated with applying the autonomous and non-autonomous TCA to real video data is the computation of loss between the TCA state $\mathbf{X}(t)$ and the frame $\mathbf{S}(t)$. The most straightforward approach is to generate “predicted frame”

In order to understand the efficacy of TCA, we will also aim to compare the state prediction performance of TCA (i.e., loss $\mathcal{L}_{pred}$) with other models, including continuum models like the one presented in (Yang et al., 2022) and frame-to-frame models discussed earlier. Since continuum models are viable on this data, it will be fascinating to understand the relative merits of TCA versus Li *et al.*’s work on Fourier Neural Operators (Li et al., 2021) from Prof. Anima Anandkumar’s group for learning a PDE characterizing the continuum dynamics of the active matter system.

5. Discussion

In addition to prediction loss comparisons, we are motivated to understand the **meaning** behind what the models we train learn. Examining the hidden states of each $x_i(t)$ over time, applying dimensionality reduction techniques to understand information flows, and comparing these to physically derived parameters (both from the statistical physics and kinematic physics perspectives) is of great interest. As mentioned in the introduction, we are not only interested in accurate models, but also in informative models that teach us about the nature of the systems we study – and perhaps about emergence itself.

If TCA has the highest prediction error compared with other models, and if the model is able to learn highly generalizable dynamics (e.g., predict dynamics in new light activation configurations or experimental conditions), we would consider the TCA idea corroborated. Furthermore, if we are

able to gain some fundamental insight into the nature of the system – for instance, uncover some highly descriptive metric for understanding the state of an active matter system by dissecting the hidden states of each $x_i(t)$ – we would consider the idea corroborated.

There are myriad ways for a novel method like this to be proven wrong or ineffective. One of the greatest concerns is of memory consumption: transformers are already notoriously memory hungry due to their quadratic complexity. While methods like those in (Keles et al., 2022) can reduce this, and the distance-gated attention idea is inviting, we are essentially adding gasoline to the fire by unrolling this computation through time using our dynamical formulation in Equations ?? . It is also possible that the latent variable passing through time does not add substantial value to prediction or understanding, and that frame-to-frame models end up being simpler, faster, and more performant for understanding this system.

Finally, we may achieve inconclusive results for several reasons. Our datasets are relatively small, and while one may consider each particle in the system to be a data point, the number of real experimental recordings at our disposal is small, and synthetic data only gets you so far in understanding the applicability of a model in the real world. We also may observe a high degree of variability in our models across various experimental conditions. Microfluidics is notoriously difficult to standardize, and active matter is no exception. Small changes in pH can lead to drastic changes in behavior. Finally, we may struggle to compare “prediction loss”. While the expression in Equation 4 is simple and inviting from a machine learning perspective, it can be difficult to ascertain and compare for certain classes of models. Continuum models have the advantage of operating in a continuous space, whereas TCA operates in a semi-discrete space of tokens representing particles. Moreover, it is unclear whether there will be a correspondence between the tokens and individual particles, or if the dynamics inferred by the TCA will be a “good explanation” of the **appearance** of the system, but not a “good explanation” of the **nature** of the system.

References

- Akbari, H., Yuan, L., Qian, R., Chuang, W.-H., Chang, S.-F., Cui, Y., and Gong, B. VATT: Transformers for Multimodal Self-Supervised Learning from Raw Video, Audio and Text, December 2021. Comment: Published in the 35th Conference on Neural Information Processing Systems (NeurIPS 2021).
- Arnab, A., Dehghani, M., Heigold, G., Sun, C., Lučić, M., and Schmid, C. ViViT: A Video Vi-

- sion Transformer, November 2021. Comment: ICCV 2021. Code at <https://github.com/google-research/scenic/tree/main/scenic/projects/vivit>.
- Atito, S. SiT: Self-supervised vIision Transformer. pp. 13.
- Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., and Amodei, D. Language Models are Few-Shot Learners, July 2020. Comment: 40+32 pages.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, June 2021. Comment: Fine-tuning code and pre-trained models are available at https://github.com/google-research/vision_transformer. ICLR camera-ready version with 2 small modifications: 1) Added a discussion of CLS vs GAP classifier in the appendix, 2) Fixed an error in exaFLOPs computation in Figure 5 and Table 6 (relative performance of models is basically not affected).
- Fan, H., Xiong, B., Mangalam, K., Li, Y., Yan, Z., Malik, J., and Feichtenhofer, C. Multiscale Vision Transformers, April 2021. Comment: Technical report.
- Geneva, N. and Zabarar, N. Transformers for Modeling Physical Systems. <https://arxiv.org/abs/2010.03957v6>, October 2020.
- Kalantari, S., Nazemi, E., and Masoumi, B. Emergence phenomena in self-organizing systems: a systematic literature review of concepts, researches, and future prospects. *Journal of Organizational Computing and Electronic Commerce*, 30(3):224–265, 2020. doi: 10.1080/10919392.2020.1748977. URL <https://doi.org/10.1080/10919392.2020.1748977>.
- Keles, F. D., Wijewardena, P. M., and Hegde, C. On The Computational Complexity of Self-Attention, September 2022.
- Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., and Anandkumar, A. Fourier Neural Operator for Parametric Partial Differential Equations, May 2021.
- Mordvintsev, A., Randazzo, E., Niklasson, E., and Levin, M. Growing neural cellular automata. *Distill*, 2020. doi: 10.23915/distill.00023. <https://distill.pub/2020/growing-ca>.
- Nedelec, F. and Foethke, D. Collective Langevin dynamics of flexible cytoskeletal fibers. *New Journal of Physics*, 9(11):427, November 2007. ISSN 1367-2630. doi: 10.1088/1367-2630/9/11/427.
- O’Connor, T. Emergent Properties. In Zalta, E. N. (ed.), *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, Winter 2021 edition, 2021.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. Language Models are Unsupervised Multi-task Learners. pp. 24.
- Tesfaldet, M., Nowrouzezahrai, D., and Pal, C. Attention-based Neural Cellular Automata, November 2022. Comment: NeurIPS 2022.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. Attention Is All You Need, December 2017. Comment: 15 pages, 5 figures.
- Weisstein, E. W. Cellular Automaton. <https://mathworld.wolfram.com/>.
- Yan, H., Ma, X., and Pu, Z. Learning dynamic and hierarchical traffic spatiotemporal features with transformer. *IEEE Transactions on Intelligent Transportation Systems*, 23(11):22386–22399, 2022a. doi: 10.1109/TITS.2021.3102983.
- Yan, W., Ansari, S., Lamson, A., Glaser, M. A., Blackwell, R., Betterton, M. D., and Shelley, M. Toward the cellular-scale simulation of motor-driven cytoskeletal assemblies. *eLife*, 11:e74160, May 2022b. ISSN 2050-084X. doi: 10.7554/eLife.74160.
- Yang, F., Liu, S., Lee, H. J., and Thomson, M. Dynamic Flow Control Through Active Matter Programming Language, August 2022. Comment: 9 pages, 4 figures.